Functional Requirements

Project: Real-Time IoT Data Pipeline **Team Leader:** Rana Shaaban Abdelqader Mohamed
Section: 3. Requirements Gathering

1. Purpose

This document lists the features and functionality the system must provide. Each requirement has a unique ID, a MoSCoW priority, the component owner, and a link back to the user story or use case it comes from.

Priority key: M = Must have, **S** = Should have, **C** = Could have, **W** = Won't have (this release).

2. Data Simulation (Data Capture Layer)

ID	Requirement	Priority	Owner	Source
FR-01	The system shall generate simulated sensor readings containing sensor ID, timestamp, temperature, humidity, and location	M	Mostafa Ahmed	US-01, UC-01
FR-02	The simulator shall generate a new reading every 5 seconds	M	Mostafa Ahmed	US-01, UC-01
FR-03	The simulator shall support multiple sensor IDs and locations	S	Mostafa Ahmed	US-10
FR-04	The simulator shall occasionally produce out-of-range values so anomalies and alerts can be demonstrated	M	Mostafa Ahmed	US-06, US-08
FR-05	The simulator shall follow the data schema agreed and frozen in Week 1	M	Mostafa Ahmed	Proposal 3.6 (risks)

3. Ingestion and Messaging (Transmission Layer)

ID	Requirement	Priority	Owner	Source
FR-06	The Flask API shall expose an endpoint that receives sensor readings	M	Waleed Hamouda	US-02, UC-02
FR-07	The API shall validate each reading against the schema (required fields and data types)	S	Waleed Hamouda	US-03, UC-02
FR-08	The API shall reject invalid readings and return a clear error response	S	Waleed Hamouda	US-03, UC-02
FR-09	The Kafka producer shall publish every valid reading to a Kafka topic	M	Waleed Hamouda	US-02, UC-02
FR-10	The system shall provide one Kafka topic that feeds both the batch consumer and the streaming consumer	M	Waleed Hamouda	Proposal 3.3
FR-11	The API shall log failures (validation errors, Kafka unavailable)	C	Waleed Hamouda	US-14

4. Batch Path (Processing Layer)

ID	Requirement	Priority	Owner	Source
FR-12	The batch Kafka consumer shall read readings from the topic and write them to batched files	M	Abdelrahman Sayed	US-04, UC-03
FR-13	The ETL job shall clean the data: remove duplicates, handle missing values, and fix data types	M	Shahd Safwat	US-04, UC-03
FR-14	The ETL job shall flag anomalous readings	M	Shahd Safwat	US-04, US-08
FR-15	The ETL job shall compute averages (for example per sensor over a time window)	M	Shahd Safwat	US-04

ID	Requirement	Priority	Owner	Source
FR-16	The ETL job shall load the cleaned and enriched data into a Snowflake table	M	Shahd Safwat	US-05, UC-03
FR-17	The ETL job shall exclude and log records that fail cleaning	S	Shahd Safwat	UC-03 (A1)
FR-18	The ETL job shall produce a data validation report (record counts, rejected records, completeness)	M	Shahd Safwat	Week 2 deliverable
FR-19	Batched files shall be kept until the Snowflake load succeeds, so a failed load can be retried	S	Abdelrahman Sayed	UC-03 (A2)

5. Streaming Path (Processing Layer)

ID	Requirement	Priority	Owner	Source
FR-20	The streaming consumer shall read each reading from the Kafka topic in real time	M	Mostafa Ahmed	US-06, UC-04
FR-21	The consumer shall apply threshold rules to temperature and humidity	M	Mostafa Ahmed	US-06, UC-04
FR-22	The system shall raise an alert when a threshold is breached, including sensor ID, metric, value, threshold, and timestamp	M	Mostafa Ahmed	US-06, UC-04
FR-23	Threshold values shall be configurable without changing the code	S	Mostafa Ahmed	US-11
FR-24	Every alert shall be recorded so the dashboard can display the alert history	M	Mostafa Ahmed	US-09, UC-06
FR-25	The consumer shall skip and log malformed readings without stopping	S	Mostafa Ahmed	UC-04 (A2)

6. Dashboard (Presentation Layer)

ID	Requirement	Priority	Owner	Source
FR-26	The dashboard shall read processed data from Snowflake	M	Rana Shaaban	US-07, UC-05
FR-27	The dashboard shall show temperature and humidity trends over time per sensor	M	Rana Shaaban	US-07
FR-28	The dashboard shall show anomaly counts	M	Rana Shaaban	US-08
FR-29	The dashboard shall show live alerts from the alert stream as they arrive	M	Rana Shaaban	US-06, UC-05
FR-30	The dashboard shall show the alert history with timestamp, sensor ID, metric, value, and threshold	M	Rana Shaaban	US-09, UC-06
FR-31	The dashboard shall refresh automatically without a manual page reload	M	Rana Shaaban	US-07
FR-32	The dashboard shall allow filtering by sensor ID, location, or time range	S	Rana Shaaban	US-10
FR-33	The dashboard shall show a clear message when no data is available or Snowflake is unreachable	S	Rana Shaaban	UC-05 (A1, A2)

7. Documentation and Delivery

ID	Requirement	Priority	Owner	Source
FR-34	All code and project documents shall be stored in a GitHub repository	M	All members	US-12, Proposal (course requirements)
FR-35	Each week's deliverables shall be submitted as listed in the proposal (Weeks 1 to 4)	M	All members	US-12

ID	Requirement	Priority	Owner	Source
FR-36	The team shall produce a final report and final presentation slides	M	All members	US-13
FR-37	The system shall be runnable for a live end-to-end demo	M	All members	US-13

8. MoSCoW Summary

Priority	Count	Meaning
Must have	27	Required for the project to be accepted
Should have	9	Important, included if time allows
Could have	1	Nice to have, first to drop if time is short
Won't have	5 (see below)	Explicitly excluded from this release

Won't Have (Out of Scope)

ID	Excluded Item	Reason
W-01	Real IoT hardware and device firmware	Data is simulated only
W-02	Machine learning or predictive models	Outside project scope
W-03	Production-grade security	Outside project scope (see Non-Functional Requirements)
W-04	Production-grade scaling	Outside project scope
W-05	Cost optimization	Outside project scope

9. Requirements to Objectives Traceability

Proposal Objective	Related Requirements
Simulate sensor data every 5 seconds	FR-01 to FR-05
Ingest through Flask API and Kafka producer	FR-06 to FR-11
Batch ETL into Snowflake	FR-12 to FR-19
Streaming pipeline with real-time alerts	FR-20 to FR-25
Live dashboard	FR-26 to FR-33
Document the system and present findings	FR-34 to FR-37